



UNIVERSITY OF THE PUNJAB

Sixth Semester - 2017
Examination: B.S. 4 Years Programme

Roll No.

PAPER: Advanced Experimental Design (Theory)
Course Code: STAT-310

TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.
SHORT QUESTIONS

Q2. Differentiate between the following terms:

(9. 6. 5)

- Fixed Effect, Random Effect and Mixed Effect Model
- Complete and Partial Confounding
- Balanced Incomplete Block Design and Partially Balanced Incomplete Block Design

SUBJECTIVE

Q3. Given the following ANOVA table:

(08)

SOV	df	Mean Square
A	2	5341.86
B	2	19558.36
AB	4	2403.44
Error	27	675.21
Total	35	

Assuming completely randomized model, interpret the effects of A, B and AB, when

- The levels of both factors are fixed
- The levels of both factors are random
- Levels of A are fixed but levels of B are random

Q4. Suppose that 4 factors, temperature (A), pressure (B), concentration of reactant (c), and stirring rate (D) are studied in a pilot plant to determine their effect on product filtration rate. Suppose now that the 24 = 16 treatment combinations cannot all be run on the same day. The experimenter can run 8 treatment combinations in one day, so a 24 design confounded in 2 blocks seems appropriate. It is logical to confound the highest interaction ABCD with blocks. (08)

Block 1	Block 2
(1) = 45	a = 71
ab = 65	b = 48
ac = 60	c = 68
bc = 80	d = 43
ad = 100	abc = 65
bd = 45	bcd = 70
cd = 75	acd = 86
abcd = 96	abd = 104

Q5. Complete the following table

(06)

S.O.V	d.f.	SS	MS	F-Ratio
Block	3	28.44	-	-
A	-	4.92	1.64	-
Error (a)	-	-	-	-
Subtotal	-	40.85	-	-
B	-	29.30	-	-
AB	9	-	-	-
Error (b)	-	86.65	-	-

Complete the ANOVA table and test the significance of difference among the treatment means.

Q6. An engineer is studying the mileage performance of 5 types of gasoline additive. In the road test he wishes to use cars as blocks, however, because of a time constraint, he must use an incomplete block design. He runs the balanced design with the 5 blocks that follow. Analyze the data from this experiment and draw conclusions. (08)

Additive	Car				
	1	2	3	4	5
1	-	17	14	13	12
2	14	14	-	13	10
3	12	-	13	12	11
4	13	11	11	12	-
5	11	12	10	-	10



UNIVERSITY OF THE PUNJAB

Sixth Semester - 2018

Examination: B.S. 4 Years Programme

Roll No.

PAPER: Advanced Experimental Design (Theory)

TIME ALLOWED: 2 Hrs. & 45 Mins.

Course Code: STAT-310 Part - II

MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

SHORT QUESTIONS

Q2. Define the following:

(5x4=20)

- Factorial Design
- Confounding
- Youden Square Design
- Incomplete Block Design

SUBJECTIVE

Q3. Suppose a study is performed to determine the effect of 3 factors (A, B, C). Suppose that each batch of raw material is only large enough to test only 4 treatment combinations. Thus, each replicate of 2^3 design must be run in 2 blocks. Two replicates are run with ABC confounded in replicate-I and AB confounded in replicate-II. The data is as follows: (08)

Replicate-I		Replicate-II	
ABC Confounded		AB Confounded	
Block 1	Block 2	Block 1	Block 2
(1) = -3	a = 0	(1) = -1	a = 1
ab = 2	b = -1	c = 0	b = 0
ac = 2	c = -1	ab = 3	ac = 1
bc = 1	abc = 6	abc = 5	bc = 1

Analyze the experiment.

Q4. Consider the 2^5 design in four blocks of eight run each with ACDE and BCE as independent effects chosen to be confounded with blocks. Generate the design. Find the other effects confounded with blocks. (06)

Q5. Complete and analyze the following ANOVA table: (08)

S.O.V	d.f	SS	MS	F-Ratio
Varieties (V)	2	0.1781	-	-
Blocks	-	4.1499	-	-
Error (a)	10	1.3622	-	-
Dates (D)	-	1.9625	-	-
V X D	-	0.2105	0.0351	-
Error (b)	-	1.2583	-	-

Q6. Consider the following partially balanced incomplete block design (08)

Blocks	Treatment Combinations		
1	1	2	3
2	3	4	5
3	2	5	6
4	1	2	4
5	3	4	6
6	1	5	6

Verify the following relationships among the parameters

$$p_{11}^1 + p_{12}^2 = n_1$$

$$n_1 p_{12}^1 = n_2 p_{11}^2$$

$$p_{21}^1 + p_{22}^2 = n_2$$

$$n_1 p_{11}^1 = n_2 p_{12}^2$$

$$p_{11}^1 + p_{12}^1 = n_1 - 1$$

$$n_1 \lambda_1 + n_2 \lambda_2 = r(k-1)$$

$$p_{21}^1 + p_{22}^2 = n_2 - 1$$

$$n_1 + n_2 = a - 1$$



ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

SHORT QUESTIONS

Q2. Define the following:

(5*4)

- i. Advantages and Disadvantages of Factorial Design
- ii. iii. Fractional Replication
- iii. Quasi Latin Squares
- iv. iv. Split-Split Plot Designs

LONG QUESTIONS

Q3. Complete ANOVA table for the following factorial experiment. ABC is completely confounded. (08)

Replicate I		Replicate II	
Block I	Block 2	Block I	Block 2
b=4	(1)=10	a=6	bc=6
a=6	ab=7	abc=9	ac=8
c=9	bc=8	c=8	(1)=11
abc=5	ac=6	b=7	ab=8

Q4. In a split plot design the whole plot treatment (A) were applied to 3 blocks and sub-plot treatment (B) were then applied to split plot. The total SS, Block SS, whole Plot Error SS are 822.97, 77.55 and 36.28 respectively. The sum for 3 blocks are given in the following table:

	B ₁	B ₂	B ₃	B ₄
A ₁	89	104	118	117
A ₂	100	117	119	126
A ₃	92	92	104	121

Complete the ANOVA and draw your conclusions. Also Compute the standard error for the

- i) difference between two 'A' treatment means
- ii) difference between two 'B' treatment means

(08)

Q5. Consider the 2^{4-1} fractional factorial with generator I = BCDE = ACDF = ABCG = ABDH. Find the aliases of C. (06)

Q6. Data on screen colour difference on a television tube measured in degrees Kelvin are to be compared for Four operators. On a given day only three operators can be used in the experiment. A balanced incomplete block design gave results as follows: (08)

Operators	Days			
	Monday	Tuesday	Wednesday	Thursday
A	173	174	-	171
B	-	175	167	172
C	183	175	168	-
D	175	-	172	175

Do a complete analysis of these data and discuss your findings with regard to difference between operators.



UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Sixth Semester – Spring 2022

Roll No.

Paper: Advanced Experimental Design (Theory)
Course Code: STAT-310

Time: 3 Hrs. Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Write a short note on the following.

(6x5=30)

- I. Factorial Experiments
- II. Fractional replication
- III. Confounding

- IV. Design resolution
- V. Youdon Square
- VI. PBIBD

Answer the following Questions.

(3x10=30)

Q2. Complete the ANOVA table for the following factorial experiment. ABC is completely confounded with the replicates.

Replicate I		Replicate II	
Blocks I	Blocks II	Blocks I	Blocks II
b=4	(1)=10	a=6	bc=6
a=6	ab=7	abc=9	ac=8
c=9	bc=8	c=8	(1)=11
abc=5	ac=6	b=7	ab=8

Q3. Consider a 2^{8-4} the fractional factorial with generator $I=BCDE=ACDE=ABCG=ABDH$. Find the aliases of C.

Q4. Consider the following partially balanced Incomplete Block design (PBIBD)

Blocks					
I	II	III	IV	V	VI
1	3	2	1	3	1
2	4	5	2	4	5
3	5	6	4	6	6

Verify:

$$(i) p_{11}^1 + p_{12}^2 = n_1$$

$$(ii) p_{11}^1 + p_{12}^1 = n_1 - 1$$

$$(iii) n_1 p_{12}^1 = n_2 p_{11}^2$$

$$(iv) n_1 p_{22}^1 = n_2 p_{12}^2$$

$$(v) p_{21}^2 + p_{22}^2 = n_2 - 1$$

$$(vi) n_1 + n_2 = a - 1$$

$$(vii) n_1 \lambda_1 + n_2 \lambda_2 = r(k-1)$$

$$(viii) p_{21}^1 + p_{22}^1 = n_2$$

**THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED****Q.1. Write a short note on the following.****(6x5=30)**

- I. Resolution IV Designs
- II. Yates Method for Computing the Contrast and Sum of Squares of an Effect
- III. Contrast Method for Constructing the Confounding Blocks
- IV. Allases
- V. Confounding 2^4 Design in 2^2 Blocks
- VI. Youden Square Design

Answer the following questions.**(3x10=30)****PART – II (Long Essay Questions)**

- Q2.** The percentage of hardwood concentration in raw pulp, the vat pressure, and the cooking time of the pulp are being investigated for their effects on the strength of paper. Three levels of hardwood concentration, three levels of pressure, and two cooking times are selected. A factorial experiment with two replicates is conducted, and the following data are obtained:

Percentage of Hardwood Concentration	Cooking Time 3.0 Hours			Cooking Time 4.0 Hours		
	Pressure			Pressure		
	400	500	650	400	500	650
2	196.6	197.7	199.8	198.4	199.6	200.6
	196.0	196.0	199.4	198.6	200.4	200.9
4	198.5	196.0	198.4	197.5	198.7	199.6
	197.2	196.9	197.6	198.1	198.0	199.0
8	197.5	195.6	197.4	197.6	197.0	198.5
	196.6	196.2	198.1	198.4	197.8	199.8

Analyze the data and draw conclusions.

(10)

- Q3.** Construct a 2^3 design with ABC confounded in the first two replicates and BC confounded in the third. Outline the analysis of variance and comment on the information obtained.

(10)

- Q4.** Seven different hardwood concentrations are being studied to determine their effect on the strength of the paper produced. However, the pilot plant can only produce three runs each day. As days may differ, the analyst uses the balanced incomplete block design that follows. Analyze the data.

(10)

Hardwood Concentration (%)	Days						
	1	2	3	4	5	6	7
2	11.4	-	-	-	12.0	-	11.7
4	12.6	12.0	-	-	-	11.9	-
6	-	13.7	11.7	-	-	-	13.4
8	14.1	-	12.9	14.9	-	-	-
10	-	14.5	-	15.0	14.3	-	-
12	-	-	12.0	-	11.8	12.3	-
14	-	-	-	13.6	-	13.0	12.7